```
  ___  ____  ____  ____  ____(R)
 /__    /   ____/   /   ____/
___/   /   /___/   /   /___/
  Statistics/Data analysis
```

```
1 . do "C:\Users\fahim\AppData\Local\Temp\STD962c_000000.tmp"

2 .
3 . /* StrokeVolume-with interaction */
4 . destring StrokeVolume, replace force
  StrokeVolume already numeric; no replace

5 .
6 . xtset Number TimeNumerical

  Panel variable: Number (strongly balanced)
   Time variable: TimeNumerical, 0 to 74, but with gaps
           Delta: 1 unit

7 . xtmixed StrokeVolume GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) r
  Note: time gaps exist in the estimation data

  Obtaining starting values by EM:

  Performing gradient-based optimization:

  Iteration 0:   log restricted-likelihood = -175.97011  (not concave)
  Iteration 1:   log restricted-likelihood = -175.97009  (not concave)
  Iteration 2:   log restricted-likelihood = -175.93549  (backed up)
  Iteration 3:   log restricted-likelihood =  -175.9083
  Iteration 4:   log restricted-likelihood = -175.88715
  Iteration 5:   log restricted-likelihood = -175.88668
  Iteration 6:   log restricted-likelihood = -175.88668

  Computing standard errors:

  Mixed-effects REML regression                  Number of obs     =         52
  Group variable: Number                         Number of groups  =         18
                                                 Obs per group:
                                                               min =          2
                                                               avg =        2.9
                                                               max =          3
                                                 Wald chi2(10)     =      26.48
  Log restricted-likelihood = -175.88668         Prob > chi2       =     0.0031
```

StrokeVolume	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
GroupNumerical						
1	7.95936	8.418308	0.95	0.344	-8.54022	24.45894
2	-6.914528	8.59265	-0.80	0.421	-23.75581	9.926757
TimeNumerical						
30	-5.989356	5.86487	-1.02	0.307	-17.48429	5.505578
74	11.27486	6.435869	1.75	0.080	-1.339215	23.88893
GroupNumerical#TimeNumerical						
1 30	-.803305	8.574284	-0.09	0.925	-17.60859	16.00198
1 74	-2.176312	9.101693	-0.24	0.811	-20.0153	15.66268
2 30	2.278391	8.294178	0.27	0.784	-13.9779	18.53468
2 74	-19.64481	9.395425	-2.09	0.037	-38.05951	-1.230118
Sex	-.4092597	6.138959	-0.07	0.947	-12.4414	11.62288
weight	1.504259	.5445855	2.76	0.006	.4368908	2.571627
_cons	-74.10714	42.82242	-1.73	0.084	-158.0375	9.823255

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	**84.45409**	**70.29656**	**16.52384**	**431.6486**
Residual: AR(1)				
rho	**-.9447892**	**.0755781**	**-.9964099**	**-.3810543**
var(e)	**126.1478**	**53.60567**	**54.84877**	**290.1297**

LR test vs. linear model: chi2(**2**) = **6.02**   Prob > chi2 = **0.0493**

Note: LR test is conservative and provided only for reference.

8 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
StrokeVolume			
TimeNumerical@GroupNumerical			
0	**2**	**7.76**	**0.0206**
1	**2**	**5.85**	**0.0537**
2	**2**	**1.50**	**0.4722**
Joint	**6**	**15.11**	**0.0194**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
StrokeVolume						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-5.989356**	**5.86487**	**-1.02**	**0.307**	**-17.48429**	**5.505578**
(30 vs base) 1	**-6.792661**	**6.25473**	**-1.09**	**0.277**	**-19.05171**	**5.466385**
(30 vs base) 2	**-3.710964**	**5.86487**	**-0.63**	**0.527**	**-15.2059**	**7.783969**
(74 vs base) 0	**11.27486**	**6.435869**	**1.75**	**0.080**	**-1.339215**	**23.88893**
(74 vs base) 1	**9.098544**	**6.435869**	**1.41**	**0.157**	**-3.515527**	**21.71261**
(74 vs base) 2	**-8.369956**	**6.844969**	**-1.22**	**0.221**	**-21.78585**	**5.045936**

9 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
StrokeVolume			
GroupNumerical@TimeNumerical			
0	**2**	**2.86**	**0.2390**
30	**2**	**1.72**	**0.4228**
74	**2**	**13.66**	**0.0011**
Joint	**6**	**14.84**	**0.0215**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
StrokeVolume						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**7.95936**	**8.418308**	**0.95**	**0.344**	**-8.54022**	**24.45894**
(1 vs base) 30	**7.156055**	**8.68692**	**0.82**	**0.410**	**-9.869995**	**24.18211**
(1 vs base) 74	**5.783048**	**8.418308**	**0.69**	**0.492**	**-10.71653**	**22.28263**
(2 vs base) 0	**-6.914528**	**8.59265**	**-0.80**	**0.421**	**-23.75581**	**9.926757**
(2 vs base) 30	**-4.636137**	**8.59265**	**-0.54**	**0.590**	**-21.47742**	**12.20515**
(2 vs base) 74	**-26.55934**	**8.976099**	**-2.96**	**0.003**	**-44.15217**	**-8.966509**

```
10 .
11 . /* Repeat with natural log */
12 . generate logStrokeVolume = ln(StrokeVolume) /*natural log transform*/
   (2 missing values generated)

13 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

14 . xtmixed logStrokeVolume GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)
   Note: time gaps exist in the estimation data

   Obtaining starting values by EM:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -14.035373  (not concave)
   Iteration 1:   log restricted-likelihood = -14.035365  (backed up)
   Iteration 2:   log restricted-likelihood = -14.035362  (not concave)
   Iteration 3:   log restricted-likelihood = -14.032339  (not concave)
   Iteration 4:   log restricted-likelihood =  -13.90358
   Iteration 5:   log restricted-likelihood = -13.871918
   Iteration 6:   log restricted-likelihood = -13.718464
   Iteration 7:   log restricted-likelihood = -13.717707
   Iteration 8:   log restricted-likelihood = -13.717705

   Computing standard errors:

   Mixed-effects REML regression                   Number of obs     =         52
   Group variable: Number                          Number of groups  =         18
                                                   Obs per group:
                                                                 min =          2
                                                                 avg =        2.9
                                                                 max =          3
                                                   Wald chi2(10)     =      25.21
   Log restricted-likelihood = -13.717705          Prob > chi2       =     0.0050
```

logStrokeVolume	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
GroupNumerical						
1	**.1244023**	**.1622683**	**0.77**	**0.443**	**-.1936377**	**.4424423**
2	**-.1477532**	**.1656139**	**-0.89**	**0.372**	**-.4723505**	**.1768442**
TimeNumerical						
30	**-.1556222**	**.1072664**	**-1.45**	**0.147**	**-.3658606**	**.0546161**
74	**.2384214**	**.130613**	**1.83**	**0.068**	**-.0175753**	**.4944181**
GroupNumerical#TimeNumerical						
1 30	**.0438122**	**.1568534**	**0.28**	**0.780**	**-.2636148**	**.3512392**
1 74	**-.1321561**	**.1847147**	**-0.72**	**0.474**	**-.4941901**	**.229878**
2 30	**.1034498**	**.1516976**	**0.68**	**0.495**	**-.1938721**	**.4007717**
2 74	**-.4124816**	**.1903098**	**-2.17**	**0.030**	**-.785482**	**-.0394812**
Sex	**-.0082204**	**.1177827**	**-0.07**	**0.944**	**-.2390701**	**.2226294**
weight	**.0251679**	**.0104737**	**2.40**	**0.016**	**.0046399**	**.0456959**
_cons	**1.787316**	**.8236661**	**2.17**	**0.030**	**.1729599**	**3.401672**

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	**.0214001**	**.0412441**	**.0004897**	**.9352473**
Residual: AR(1)				
rho	**.9693353**	**.0376644**	**.6956125**	**.9973024**
var(e)	**.0568525**	**.0379143**	**.0153848**	**.2100904**

LR test vs. linear model: chi2(**2**) = **7.34**   Prob > chi2 = **0.0255**

Note: LR test is conservative and provided only for reference.

15 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logStrokeVolume			
TimeNumerical@GroupNumerical			
0	**2**	**11.11**	**0.0039**
1	**2**	**3.11**	**0.2114**
2	**2**	**1.60**	**0.4489**
Joint	**6**	**15.82**	**0.0148**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logStrokeVolume						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-.1556222**	**.1072664**	**-1.45**	**0.147**	**-.3658606**	**.0546161**
(30 vs base) 1	**-.11181**	**.1144417**	**-0.98**	**0.329**	**-.3361116**	**.1124915**
(30 vs base) 2	**-.0521725**	**.1072664**	**-0.49**	**0.627**	**-.2624108**	**.1580659**
(74 vs base) 0	**.2384214**	**.130613**	**1.83**	**0.068**	**-.0175753**	**.4944181**
(74 vs base) 1	**.1062653**	**.130613**	**0.81**	**0.416**	**-.1497314**	**.3622621**
(74 vs base) 2	**-.1740602**	**.1384127**	**-1.26**	**0.209**	**-.4453441**	**.0972237**

```
16 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logStrokeVolume			
GroupNumerical@TimeNumerical			
0	**2**	**2.57**	**0.2764**
30	**2**	**1.67**	**0.4334**
74	**2**	**12.76**	**0.0017**
Joint	**6**	**15.51**	**0.0167**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logStrokeVolume						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**.1244023**	**.1622683**	**0.77**	**0.443**	**-.1936377**	**.4424423**
(1 vs base) 30	**.1682145**	**.1669874**	**1.01**	**0.314**	**-.1590747**	**.4955038**
(1 vs base) 74	**-.0077537**	**.1622683**	**-0.05**	**0.962**	**-.3257937**	**.3102863**
(2 vs base) 0	**-.1477532**	**.1656139**	**-0.89**	**0.372**	**-.4723505**	**.1768442**
(2 vs base) 30	**-.0443034**	**.1656139**	**-0.27**	**0.789**	**-.3689007**	**.280294**
(2 vs base) 74	**-.5602348**	**.1734049**	**-3.23**	**0.001**	**-.9001022**	**-.2203673**

```
17 .
18 .
19 . /* StrokeVolume -without interaction */
20 . xtset Number TimeNumerical
```

Panel variable: **Number** (strongly balanced)
 Time variable: **TimeNumerical, 0** to **74,** but with gaps
         Delta: **1 unit**

```
21 . xtmixed StrokeVolume Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0:   log restricted-likelihood = -191.78079
Iteration 1:   log restricted-likelihood = -191.78078

Computing standard errors:

Mixed-effects REML regression                   Number of obs     =         52
Group variable: Number                          Number of groups  =         18
                                                Obs per group:
                                                              min =          2
                                                              avg =        2.9
                                                              max =          3
                                                Wald chi2(6)      =      16.27
Log restricted-likelihood = -191.78078          Prob > chi2       =     0.0124
```

StrokeVolume	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
Sex	-.584318	6.442828	-0.09	0.928	-13.21203	12.04339
weight	1.361904	.5696367	2.39	0.017	.245437	2.478372
GroupNumerical						
1	6.947628	7.103955	0.98	0.328	-6.975868	20.87112
2	-11.37597	7.356943	-1.55	0.122	-25.79531	3.043374
TimeNumerical						
30	-5.302655	3.786207	-1.40	0.161	-12.72348	2.118175
74	4.830274	3.7872	1.28	0.202	-2.592501	12.25305
_cons	-60.88349	44.64483	-1.36	0.173	-148.3858	26.61877

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	106.3367	63.14929	33.20379	340.5486
var(Residual)	124.1181	31.6165	75.33714	204.4848

```
LR test vs. linear model: chibar2(01) = 6.41          Prob >= chibar2 = 0.0057

22 .
23 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: asbalanced
```

	df	chi2	P>chi2
StrokeVolume			
TimeNumerical@GroupNumerical			
0	**2**	**6.89**	**0.0319**
1	**2**	**6.89**	**0.0319**
2	**2**	**6.89**	**0.0319**
Joint	**2**	**6.89**	**0.0319**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
StrokeVolume						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-5.302655**	**3.786207**	**-1.40**	**0.161**	**-12.72348**	**2.118175**
(30 vs base) 1	**-5.302655**	**3.786207**	**-1.40**	**0.161**	**-12.72348**	**2.118175**
(30 vs base) 2	**-5.302655**	**3.786207**	**-1.40**	**0.161**	**-12.72348**	**2.118175**
(74 vs base) 0	**4.830274**	**3.7872**	**1.28**	**0.202**	**-2.592501**	**12.25305**
(74 vs base) 1	**4.830274**	**3.7872**	**1.28**	**0.202**	**-2.592501**	**12.25305**
(74 vs base) 2	**4.830274**	**3.7872**	**1.28**	**0.202**	**-2.592501**	**12.25305**

```
24 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
StrokeVolume			
GroupNumerical@TimeNumerical			
0	**2**	**5.78**	**0.0556**
30	**2**	**5.78**	**0.0556**
74	**2**	**5.78**	**0.0556**
Joint	**2**	**5.78**	**0.0556**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
StrokeVolume						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**6.947628**	**7.103955**	**0.98**	**0.328**	**-6.975868**	**20.87112**
(1 vs base) 30	**6.947628**	**7.103955**	**0.98**	**0.328**	**-6.975868**	**20.87112**
(1 vs base) 74	**6.947628**	**7.103955**	**0.98**	**0.328**	**-6.975868**	**20.87112**
(2 vs base) 0	**-11.37597**	**7.356943**	**-1.55**	**0.122**	**-25.79531**	**3.043374**
(2 vs base) 30	**-11.37597**	**7.356943**	**-1.55**	**0.122**	**-25.79531**	**3.043374**
(2 vs base) 74	**-11.37597**	**7.356943**	**-1.55**	**0.122**	**-25.79531**	**3.043374**

```
25 .
26 . /* Repeat with natural log */
27 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

28 . xtmixed logStrokeVolume  Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

   Performing EM optimization:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood =  -14.58248
   Iteration 1:   log restricted-likelihood = -14.582472
   Iteration 2:   log restricted-likelihood = -14.582472

   Computing standard errors:

   Mixed-effects REML regression                  Number of obs     =         52
   Group variable: Number                         Number of groups  =         18
                                                  Obs per group:
                                                                min =          2
                                                                avg =        2.9
                                                                max =          3
                                                  Wald chi2(6)      =      12.45
   Log restricted-likelihood = -14.582472         Prob > chi2       =     0.0527
```

logStrokeVolume	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
Sex	-.0125584	.1274487	-0.10	0.922	-.2623533	.2372364
weight	.0216621	.0112653	1.92	0.054	-.0004176	.0437417
GroupNumerical						
1	.0917607	.1405068	0.65	0.514	-.1836275	.3671489
2	-.2193326	.1454888	-1.51	0.132	-.5044853	.0658202
TimeNumerical						
30	-.1044698	.0733752	-1.42	0.155	-.2482826	.039343
74	.0760362	.0733938	1.04	0.300	-.067813	.2198854
_cons	2.102803	.8828882	2.38	0.017	.3723735	3.833232

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	.0422688	.0245605	.0135339	.1320126
var(Residual)	.0466073	.011855	.0283102	.07673

```
   LR test vs. linear model: chibar2(01) = 6.94          Prob >= chibar2 = 0.0042
```

```
29 .
30 . contrast TimeNumerical@GroupNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logStrokeVolume			
TimeNumerical@GroupNumerical			
0	**2**	**5.87**	**0.0531**
1	**2**	**5.87**	**0.0531**
2	**2**	**5.87**	**0.0531**
Joint	**2**	**5.87**	**0.0531**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logStrokeVolume						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-.1044698**	**.0733752**	**-1.42**	**0.155**	**-.2482826**	**.039343**
(30 vs base) 1	**-.1044698**	**.0733752**	**-1.42**	**0.155**	**-.2482826**	**.039343**
(30 vs base) 2	**-.1044698**	**.0733752**	**-1.42**	**0.155**	**-.2482826**	**.039343**
(74 vs base) 0	**.0760362**	**.0733938**	**1.04**	**0.300**	**-.067813**	**.2198854**
(74 vs base) 1	**.0760362**	**.0733938**	**1.04**	**0.300**	**-.067813**	**.2198854**
(74 vs base) 2	**.0760362**	**.0733938**	**1.04**	**0.300**	**-.067813**	**.2198854**

```
31 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logStrokeVolume			
GroupNumerical@TimeNumerical			
0	**2**	**4.41**	**0.1104**
30	**2**	**4.41**	**0.1104**
74	**2**	**4.41**	**0.1104**
Joint	**2**	**4.41**	**0.1104**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logStrokeVolume						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**.0917607**	**.1405068**	**0.65**	**0.514**	**-.1836275**	**.3671489**
(1 vs base) 30	**.0917607**	**.1405068**	**0.65**	**0.514**	**-.1836275**	**.3671489**
(1 vs base) 74	**.0917607**	**.1405068**	**0.65**	**0.514**	**-.1836275**	**.3671489**
(2 vs base) 0	**-.2193326**	**.1454888**	**-1.51**	**0.132**	**-.5044853**	**.0658202**
(2 vs base) 30	**-.2193326**	**.1454888**	**-1.51**	**0.132**	**-.5044853**	**.0658202**
(2 vs base) 74	**-.2193326**	**.1454888**	**-1.51**	**0.132**	**-.5044853**	**.0658202**

```
32 .
33 .
34 .
35 . /* StrokeWork-with interaction */
36 . destring StrokeWork, replace force
   StrokeWork already numeric; no replace

37 .
38 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

39 . xtmixed StrokeWork GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) rem
   Note: time gaps exist in the estimation data

   Obtaining starting values by EM:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -359.82514  (not concave)
   Iteration 1:   log restricted-likelihood =  -359.7802
   Iteration 2:   log restricted-likelihood = -359.77901
   Iteration 3:   log restricted-likelihood = -359.77901

   Computing standard errors:

   Mixed-effects REML regression                   Number of obs     =         52
   Group variable: Number                          Number of groups  =         18
                                                   Obs per group:
                                                                 min =          2
                                                                 avg =        2.9
                                                                 max =          3
                                                   Wald chi2(10)     =     224.03
   Log restricted-likelihood = -359.77901          Prob > chi2       =     0.0000
```

StrokeWork	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
GroupNumerical						
1	514.1487	670.8965	0.77	0.443	-800.7843	1829.082
2	-489.1891	679.8094	-0.72	0.472	-1821.591	843.2128
TimeNumerical						
30	-1849.289	635.7204	-2.91	0.004	-3095.279	-603.3002
74	925.639	635.7204	1.46	0.145	-320.3501	2171.628
GroupNumerical#TimeNumerical						
1 30	-74.29272	923.4968	-0.08	0.936	-1884.313	1735.728
1 74	4754.346	899.0445	5.29	0.000	2992.251	6516.441
2 30	488.5053	899.0445	0.54	0.587	-1273.589	2250.6
2 74	497.4594	924.3523	0.54	0.590	-1314.238	2309.157
Sex	8.229045	388.045	0.02	0.983	-752.3252	768.7833
weight	111.3859	34.6968	3.21	0.001	43.38146	179.3904
_cons	-5501.452	2744.603	-2.00	0.045	-10880.78	-122.1282

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	**129758.8**	**221243.4**	**4589.954**	**3668306**
Residual: AR(1)				
rho	**-2.09e-06**	**.**	**.**	**.**
var(e)	**1212421**	**1.61e-52**	**1212421**	**1212421**

LR test vs. linear model: chi2(**2**) = **0.31**   Prob > chi2 = **0.8563**

Note: LR test is conservative and provided only for reference.

40 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
StrokeWork			
TimeNumerical@GroupNumerical			
0	**2**	**19.76**	**0.0001**
1	**2**	**145.10**	**0.0000**
2	**2**	**17.25**	**0.0002**
Joint	**6**	**182.27**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
StrokeWork						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-1849.289**	**635.7204**	**-2.91**	**0.004**	**-3095.279**	**-603.3002**
(30 vs base) 1	**-1923.582**	**669.8551**	**-2.87**	**0.004**	**-3236.474**	**-610.6902**
(30 vs base) 2	**-1360.784**	**635.7204**	**-2.14**	**0.032**	**-2606.773**	**-114.795**
(74 vs base) 0	**925.639**	**635.7204**	**1.46**	**0.145**	**-320.3501**	**2171.628**
(74 vs base) 1	**5679.985**	**635.7204**	**8.93**	**0.000**	**4433.996**	**6925.974**
(74 vs base) 2	**1423.098**	**671.0341**	**2.12**	**0.034**	**107.8959**	**2738.301**

41 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
StrokeWork			
GroupNumerical@TimeNumerical			
0	**2**	**2.11**	**0.3482**
30	**2**	**0.50**	**0.7798**
74	**2**	**76.83**	**0.0000**
Joint	**6**	**77.30**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
StrokeWork						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**514.1487**	**670.8965**	**0.77**	**0.443**	**-800.7843**	**1829.082**
(1 vs base) 30	**439.856**	**702.3362**	**0.63**	**0.531**	**-936.6977**	**1816.41**
(1 vs base) 74	**5268.495**	**670.8965**	**7.85**	**0.000**	**3953.562**	**6583.427**
(2 vs base) 0	**-489.1891**	**679.8094**	**-0.72**	**0.472**	**-1821.591**	**843.2128**
(2 vs base) 30	**-.683785**	**679.8094**	**-0.00**	**0.999**	**-1333.086**	**1331.718**
(2 vs base) 74	**8.270339**	**720.052**	**0.01**	**0.991**	**-1403.006**	**1419.546**

```
42 .
43 . /* Repeat with natural log */
44 . generate logStrokeWork = ln(StrokeWork) /*natural log transform*/
   (2 missing values generated)

45 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

46 . xtmixed logStrokeWork GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical))
   Note: time gaps exist in the estimation data

   Obtaining starting values by EM:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -13.930096  (not concave)
   Iteration 1:   log restricted-likelihood = -13.927085  (not concave)
   Iteration 2:   log restricted-likelihood = -13.879174
   Iteration 3:   log restricted-likelihood =  -13.84874
   Iteration 4:   log restricted-likelihood = -13.786772
   Iteration 5:   log restricted-likelihood =  -13.78648
   Iteration 6:   log restricted-likelihood = -13.786477
   Iteration 7:   log restricted-likelihood = -13.786477

   Computing standard errors:

   Mixed-effects REML regression                   Number of obs     =         52
   Group variable: Number                          Number of groups  =         18
                                                   Obs per group:
                                                                 min =          2
                                                                 avg =        2.9
                                                                 max =          3
                                                   Wald chi2(10)     =     285.41
   Log restricted-likelihood = -13.786477          Prob > chi2       =     0.0000
```

logStrokeWork	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
GroupNumerical						
1	**.109376**	**.1462814**	**0.75**	**0.455**	**-.1773304**	**.3960824**
2	**-.0981426**	**.1483422**	**-0.66**	**0.508**	**-.388888**	**.1926029**
TimeNumerical						
30	**-.8848515**	**.1290076**	**-6.86**	**0.000**	**-1.137702**	**-.6320013**
74	**.2456759**	**.1414247**	**1.74**	**0.082**	**-.0315114**	**.5228632**
GroupNumerical#TimeNumerical						

1 30	.1804921	.1878994	0.96	0.337	-.187784	.5487683
1 74	.6788283	.2000047	3.39	0.001	.2868263	1.07083
2 30	.3362649	.1824443	1.84	0.065	-.0213193	.6938492
2 74	.1354451	.2054516	0.66	0.510	-.2672326	.5381228
Sex	.0430783	.0869183	0.50	0.620	-.1272785	.2134351
weight	.0182003	.007792	2.34	0.020	.0029282	.0334725
_cons	6.633635	.6158634	10.77	0.000	5.426565	7.840705

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	.0028959	.0237409	3.04e-10	27547.54
Residual: AR(1)				
rho	-.9444435	.0681298	-.9951888	-.4941693
var(e)	.0608891	.025933	.0264246	.1403041

LR test vs. linear model: chi2(**2**) = **0.99**   Prob > chi2 = **0.6086**

Note: LR test is conservative and provided only for reference.

47 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logStrokeWork			
TimeNumerical@GroupNumerical			
0	2	81.65	0.0000
1	2	128.56	0.0000
2	2	44.55	0.0000
Joint	6	255.01	0.0000

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logStrokeWork						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.8848515	.1290076	-6.86	0.000	-1.137702	-.6320013
(30 vs base) 1	-.7043594	.1366135	-5.16	0.000	-.9721169	-.4366019
(30 vs base) 2	-.5485866	.1290076	-4.25	0.000	-.8014368	-.2957364
(74 vs base) 0	.2456759	.1414247	1.74	0.082	-.0315114	.5228632
(74 vs base) 1	.9245042	.1414247	6.54	0.000	.6473169	1.201691
(74 vs base) 2	.381121	.1490282	2.56	0.011	.089031	.673211

```
48 . contrast GroupNumerical@TimeNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logStrokeWork			
GroupNumerical@TimeNumerical			
0	2	1.89	0.3877
30	2	4.28	0.1174
74	2	34.79	0.0000
Joint	6	39.27	0.0000

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logStrokeWork						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.109376	.1462814	0.75	0.455	-.1773304	.3960824
(1 vs base) 30	.2898682	.1528418	1.90	0.058	-.0096962	.5894325
(1 vs base) 74	.7882043	.1462814	5.39	0.000	.501498	1.074911
(2 vs base) 0	-.0981426	.1483422	-0.66	0.508	-.388888	.1926029
(2 vs base) 30	.2381224	.1483422	1.61	0.108	-.0526231	.5288678
(2 vs base) 74	.0373026	.1573045	0.24	0.813	-.2710086	.3456137

```
49 .
50 .
51 . /* StrokeWork -without interaction */
52 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

53 . xtmixed StrokeWork Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

   Performing EM optimization:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -406.80262
   Iteration 1:   log restricted-likelihood = -406.40632
   Iteration 2:   log restricted-likelihood = -406.40545
   Iteration 3:   log restricted-likelihood = -406.40545

   Computing standard errors:

   Mixed-effects REML regression                 Number of obs     =         52
   Group variable: Number                        Number of groups  =         18
                                                 Obs per group:
                                                               min =          2
                                                               avg =        2.9
                                                               max =          3
                                                 Wald chi2(6)      =      93.54
   Log restricted-likelihood = -406.40545        Prob > chi2       =     0.0000
```

StrokeWork	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]	
Sex	**-29.10589**	**481.5268**	**-0.06**	**0.952**	**-972.8812**	**914.6694**
weight	**104.6311**	**43.13989**	**2.43**	**0.015**	**20.07851**	**189.1838**
GroupNumerical						
1	**2141.571**	**534.6185**	**4.01**	**0.000**	**1093.738**	**3189.404**
2	**-89.09866**	**557.9794**	**-0.16**	**0.873**	**-1182.718**	**1004.521**
TimeNumerical						
30	**-1633.682**	**531.3952**	**-3.07**	**0.002**	**-2675.197**	**-592.1661**
74	**2724.035**	**531.7134**	**5.12**	**0.000**	**1681.895**	**3766.174**
_cons	**-5626.713**	**3387.277**	**-1.66**	**0.097**	**-12265.65**	**1012.228**

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	**3.24e-16**	**2.43e-15**	**1.32e-22**	**7.93e-10**
var(Residual)	**2464883**	**519649**	**1630591**	**3726039**

LR test vs. linear model: chibar2(01) = **2.3e-13** Prob >= chibar2 = **1.0000**

```
.
. contrast TimeNumerical@GroupNumerical , effect
```

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
StrokeWork			
TimeNumerical@GroupNumerical			
0	**2**	**66.62**	**0.0000**
1	**2**	**66.62**	**0.0000**
2	**2**	**66.62**	**0.0000**
Joint	**2**	**66.62**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf. interval]	
StrokeWork						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-1633.682**	**531.3952**	**-3.07**	**0.002**	**-2675.197**	**-592.1661**
(30 vs base) 1	**-1633.682**	**531.3952**	**-3.07**	**0.002**	**-2675.197**	**-592.1661**
(30 vs base) 2	**-1633.682**	**531.3952**	**-3.07**	**0.002**	**-2675.197**	**-592.1661**
(74 vs base) 0	**2724.035**	**531.7134**	**5.12**	**0.000**	**1681.895**	**3766.174**
(74 vs base) 1	**2724.035**	**531.7134**	**5.12**	**0.000**	**1681.895**	**3766.174**
(74 vs base) 2	**2724.035**	**531.7134**	**5.12**	**0.000**	**1681.895**	**3766.174**

```
56 . contrast GroupNumerical@TimeNumerical , effect

   Contrasts of marginal linear predictions

   Margins: asbalanced
```

	df	chi2	P>chi2
StrokeWork			
GroupNumerical@TimeNumerical			
0	2	20.28	0.0000
30	2	20.28	0.0000
74	2	20.28	0.0000
Joint	2	20.28	0.0000

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
StrokeWork						
GroupNumerical@TimeNumerical						
(1 vs base) 0	2141.571	534.6185	4.01	0.000	1093.738	3189.404
(1 vs base) 30	2141.571	534.6185	4.01	0.000	1093.738	3189.404
(1 vs base) 74	2141.571	534.6185	4.01	0.000	1093.738	3189.404
(2 vs base) 0	-89.09866	557.9794	-0.16	0.873	-1182.718	1004.521
(2 vs base) 30	-89.09866	557.9794	-0.16	0.873	-1182.718	1004.521
(2 vs base) 74	-89.09866	557.9794	-0.16	0.873	-1182.718	1004.521

```
57 .
58 . /* Repeat with natural log */
59 . xtset Number TimeNumerical

   Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
            Delta: 1 unit

60 . xtmixed logStrokeWork  Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

   Performing EM optimization:

   Performing gradient-based optimization:

   Iteration 0:   log restricted-likelihood = -19.225374
   Iteration 1:   log restricted-likelihood = -19.102587
   Iteration 2:   log restricted-likelihood = -19.102562
   Iteration 3:   log restricted-likelihood = -19.102562

   Computing standard errors:

   Mixed-effects REML regression                    Number of obs     =         52
   Group variable: Number                           Number of groups  =         18
                                                    Obs per group:
                                                                  min =          2
                                                                  avg =        2.9
                                                                  max =          3
                                                    Wald chi2(6)      =     185.58
   Log restricted-likelihood = -19.102562           Prob > chi2       =     0.0000
```

logStrokeWork	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
Sex	**.0458273**	**.0907792**	**0.50**	**0.614**	**-.1320967**	**.2237513**
weight	**.0170776**	**.0081229**	**2.10**	**0.036**	**.0011569**	**.0329983**
GroupNumerical						
1	**.4031536**	**.1007263**	**4.00**	**0.000**	**.2057337**	**.6005735**
2	**.0702349**	**.1050505**	**0.67**	**0.504**	**-.1356603**	**.2761301**
TimeNumerical						
30	**-.7035776**	**.0960798**	**-7.32**	**0.000**	**-.8918906**	**-.5152646**
74	**.5243114**	**.0961351**	**5.45**	**0.000**	**.33589**	**.7127329**
_cons	**6.568114**	**.6376596**	**10.30**	**0.000**	**5.318324**	**7.817904**

Random-effects parameters	Estimate	Std. err.	[95% conf.	interval]
Number: Identity				
var(_cons)	**.002373**	**.0144534**	**1.55e-08**	**362.9677**
var(Residual)	**.0805199**	**.0203848**	**.049024**	**.1322506**

LR test vs. linear model: chibar2(01) = **0.03** Prob >= chibar2 = **0.4333**

61 .
62 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logStrokeWork			
TimeNumerical@GroupNumerical			
0	**2**	**159.46**	**0.0000**
1	**2**	**159.46**	**0.0000**
2	**2**	**159.46**	**0.0000**
Joint	**2**	**159.46**	**0.0000**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logStrokeWork						
TimeNumerical@GroupNumerical						
(30 vs base) 0	**-.7035776**	**.0960798**	**-7.32**	**0.000**	**-.8918906**	**-.5152646**
(30 vs base) 1	**-.7035776**	**.0960798**	**-7.32**	**0.000**	**-.8918906**	**-.5152646**
(30 vs base) 2	**-.7035776**	**.0960798**	**-7.32**	**0.000**	**-.8918906**	**-.5152646**
(74 vs base) 0	**.5243114**	**.0961351**	**5.45**	**0.000**	**.33589**	**.7127329**
(74 vs base) 1	**.5243114**	**.0961351**	**5.45**	**0.000**	**.33589**	**.7127329**
(74 vs base) 2	**.5243114**	**.0961351**	**5.45**	**0.000**	**.33589**	**.7127329**

```
63 . contrast GroupNumerical@TimeNumerical , effect

   Contrasts of marginal linear predictions

   Margins: asbalanced
```

	df	chi2	P>chi2
logStrokeWork			
GroupNumerical@TimeNumerical			
0	**2**	**17.33**	**0.0002**
30	**2**	**17.33**	**0.0002**
74	**2**	**17.33**	**0.0002**
Joint	**2**	**17.33**	**0.0002**

	Contrast	Std. err.	z	P>|z|	[95% conf.	interval]
logStrokeWork						
GroupNumerical@TimeNumerical						
(1 vs base) 0	**.4031536**	**.1007263**	**4.00**	**0.000**	**.2057337**	**.6005735**
(1 vs base) 30	**.4031536**	**.1007263**	**4.00**	**0.000**	**.2057337**	**.6005735**
(1 vs base) 74	**.4031536**	**.1007263**	**4.00**	**0.000**	**.2057337**	**.6005735**
(2 vs base) 0	**.0702349**	**.1050505**	**0.67**	**0.504**	**-.1356603**	**.2761301**
(2 vs base) 30	**.0702349**	**.1050505**	**0.67**	**0.504**	**-.1356603**	**.2761301**
(2 vs base) 74	**.0702349**	**.1050505**	**0.67**	**0.504**	**-.1356603**	**.2761301**

```
64 .
   end of do-file

65 .
```